Static Website Hosting with Amazon S3 and CloudFront

A serverless approach to global content delivery with HTTPS security

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- ⚡ Enterprise-Grade Architecture
- 🌍 Global CDN Delivery
- 🔒 Production-Ready Security
- 💰 Cost-Optimized Solution

Done by Adhithyan Sivaraman T
Aspiring Cloud and DevOps Engineer

Project Objective

Design and deploy a secure, scalable, and cost-efficient static website architecture using AWS cloud-native services.

This project focuses on replacing traditional server-based hosting with a fully serverless approach, leveraging Amazon S3 for storage and Amazon CloudFront for global content delivery.

The objective is to demonstrate:

- ⚡ High-performance delivery using CDN edge caching
- 🔒 Secure access control with HTTPS and restricted S3 access (OAC)
- 🌍 Global scalability without managing servers
- 💸 Cost optimization using AWS Free Tier services

By implementing this architecture, the project showcases a production-ready solution that aligns with modern cloud and DevOps best practices.

Architecture Overview

End Users

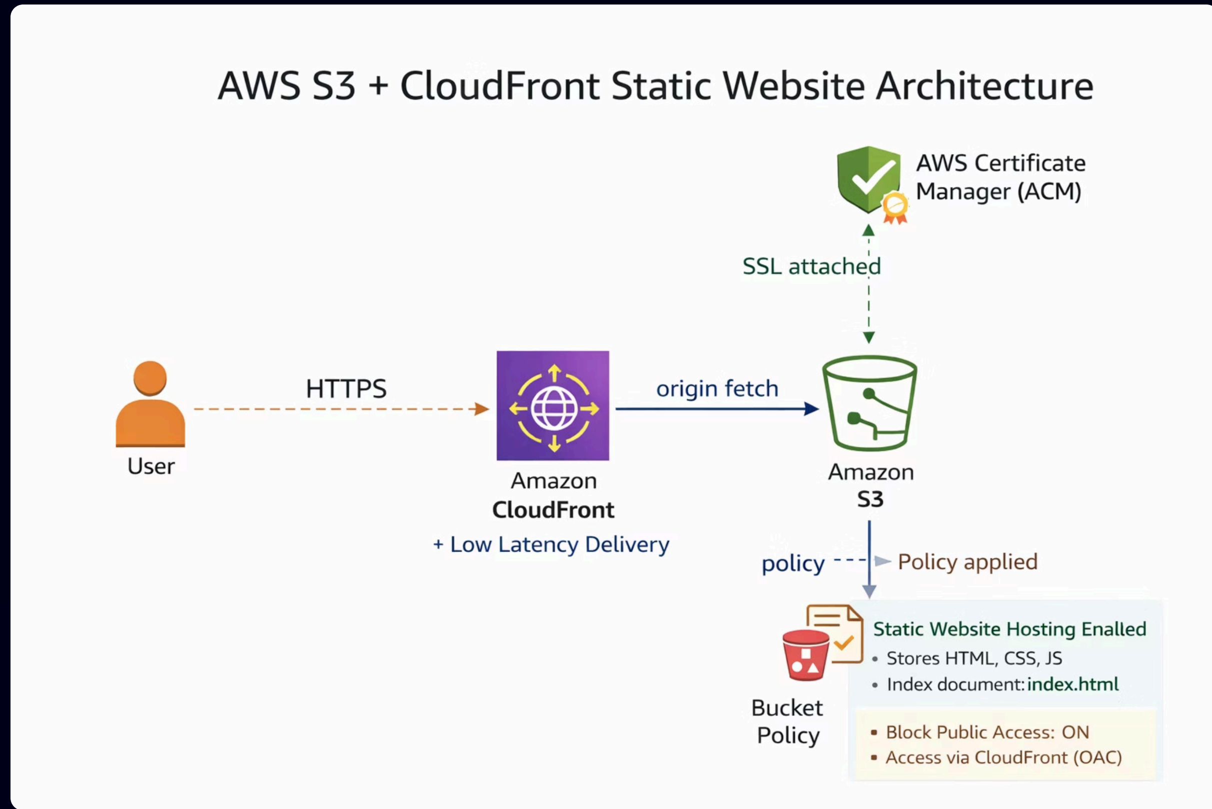
Users access the website from anywhere through a secure global delivery network.

CloudFront

Amazon CloudFront caches and delivers content through edge locations with HTTPS support.

S3 Bucket

The static website is stored in Amazon S3 as the origin source for all content.



User requests flow through Amazon CloudFront (CDN with HTTPS encryption secured by AWS Certificate Manager). CloudFront routes to the S3 bucket which stores static website files (HTML, CSS, JavaScript). The bucket policy blocks public access while allowing CloudFront Origin Access (OAC) to fetch content securely, ensuring global low-latency delivery with automatic scaling and minimal operational overhead.

Key Benefits

Serverless Hosting

No EC2 required — zero server management overhead

Global CDN

Fast loading via worldwide edge locations

Secure HTTPS

Encrypted delivery for all content

Free Tier Eligible

Cost-efficient under AWS Free Tier

Highly Scalable

Handles traffic spikes automatically

Low Maintenance

Simple to operate with minimal upkeep

Implementation Steps

1

01

S3 Bucket Setup

Created bucket, disabled block public access

2

02

File Upload

Uploaded index.html and style.css

3

03

Static Hosting

Enabled static website hosting in S3

4

04

Access Policy

Added bucket policy for public read access

5

05

CloudFront Setup

Created distribution pointing to S3 endpoint

6

06

HTTPS Enforcement

Forced HTTPS via viewer policy in CloudFront

7

07

Global Testing

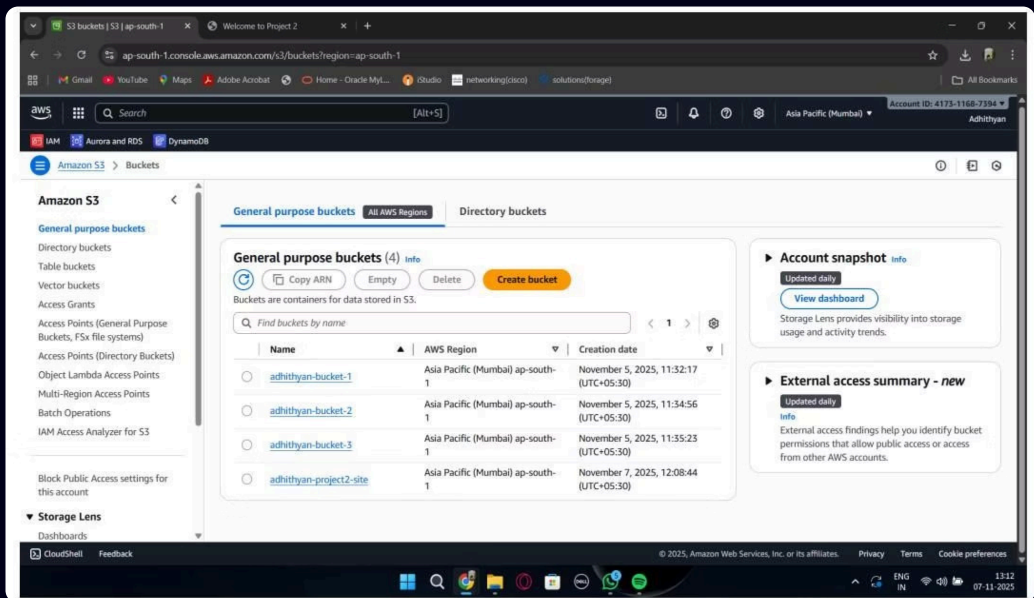
Verified delivery and performance across regions

S3 Configuration

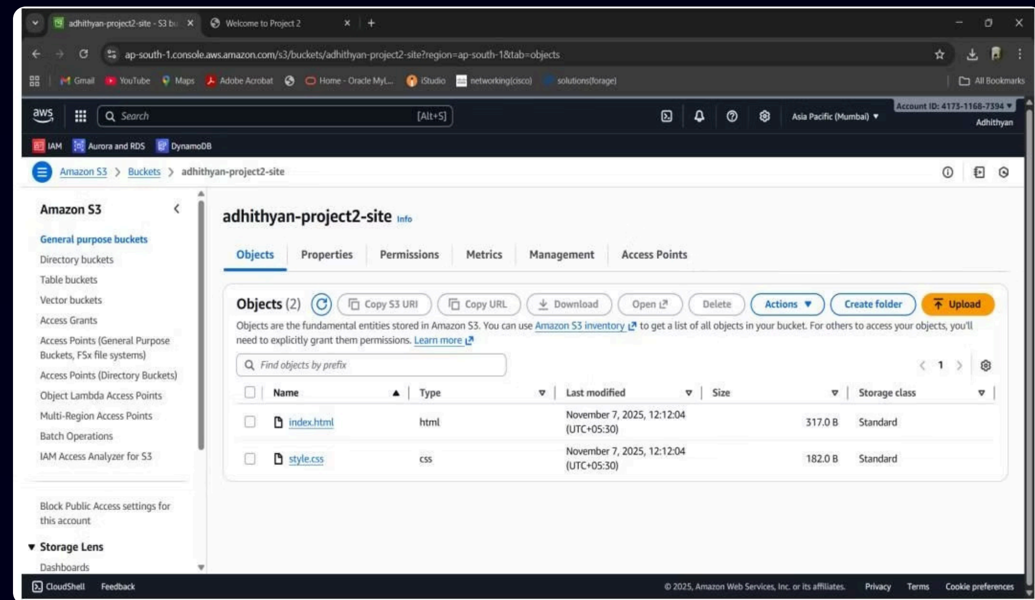
Bucket Creation & File Upload

The S3 bucket was created and configured for website hosting, then the required website files were uploaded to the bucket to prepare the site for deployment.

S3 bucket list showing created bucket



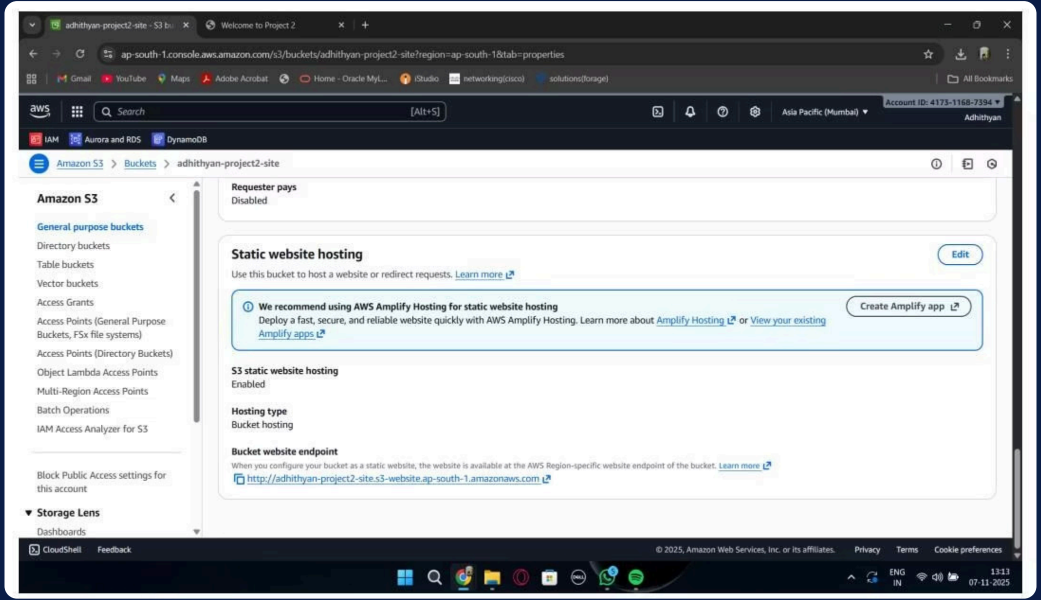
Website files uploaded to S3



Static Website Hosting & Access Policy

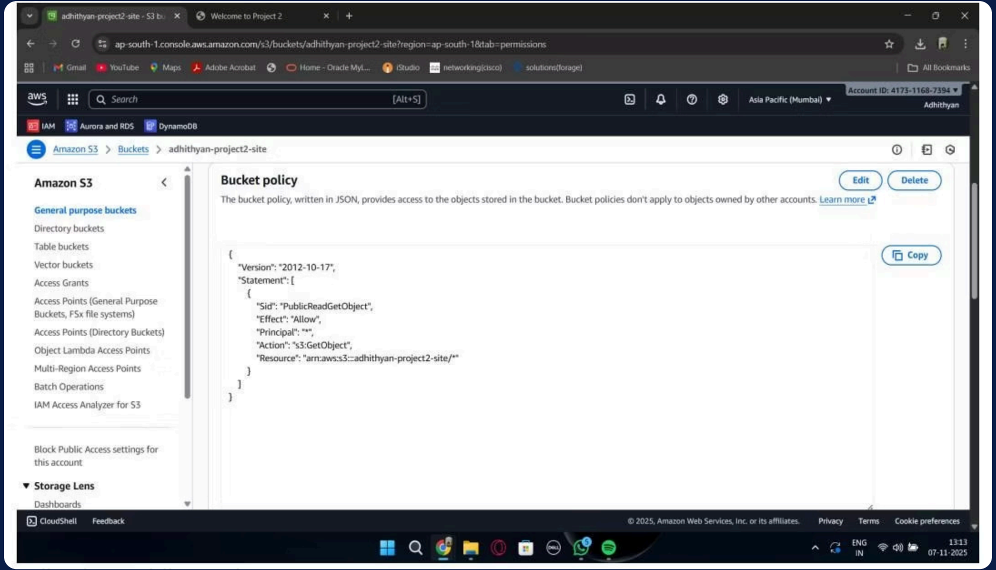
Static Hosting Enabled

S3 bucket configured to serve static website content with index document specified.



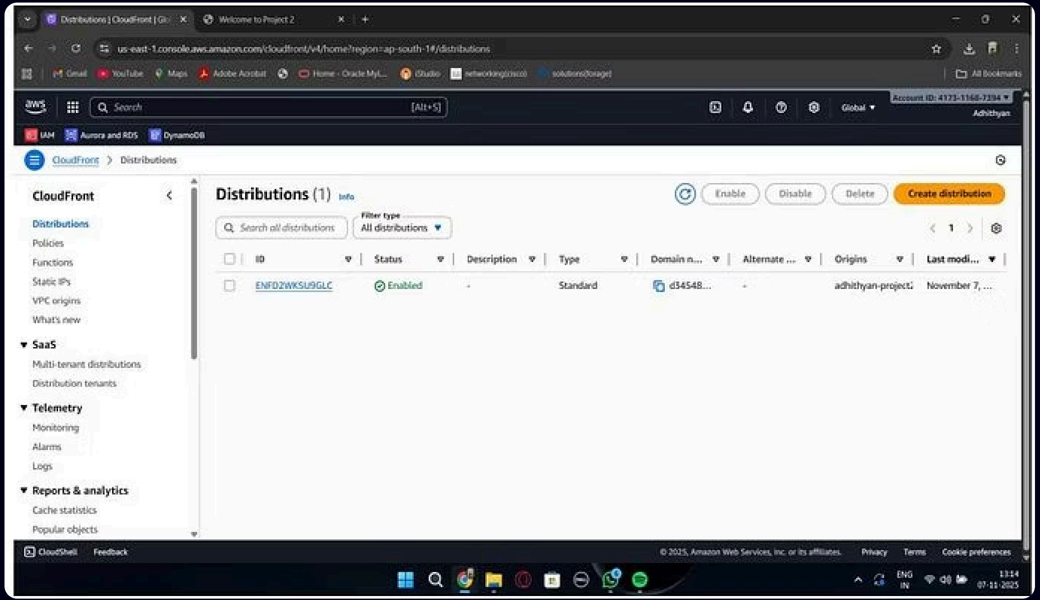
Public Read Access

Bucket policy configured to allow public read access for all objects in the bucket.

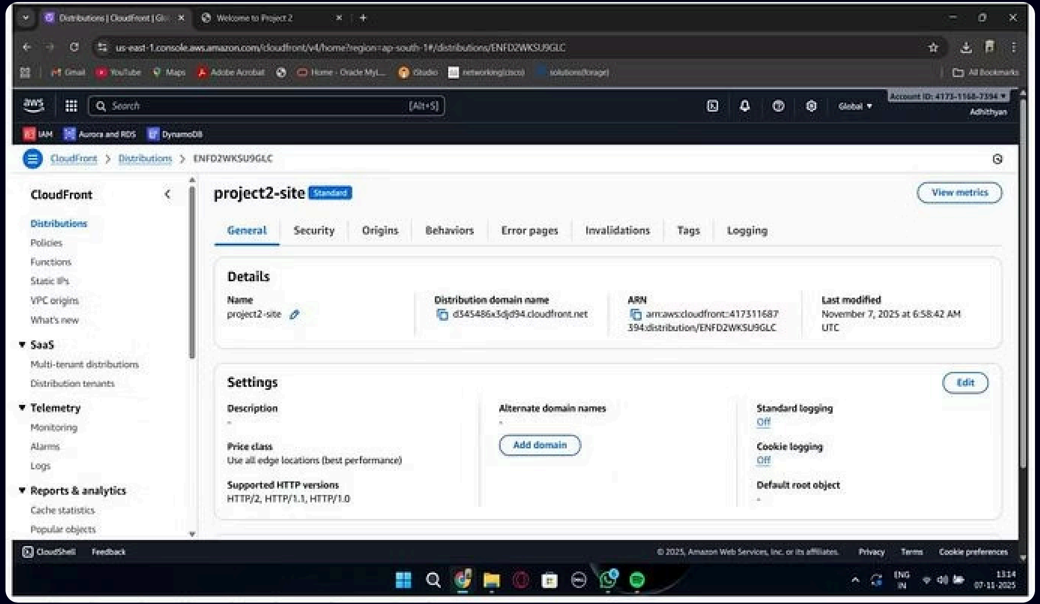


CloudFront Distribution

CloudFront CDN was configured to accelerate content delivery globally with HTTPS enforcement.



CloudFront distributions list showing enabled status

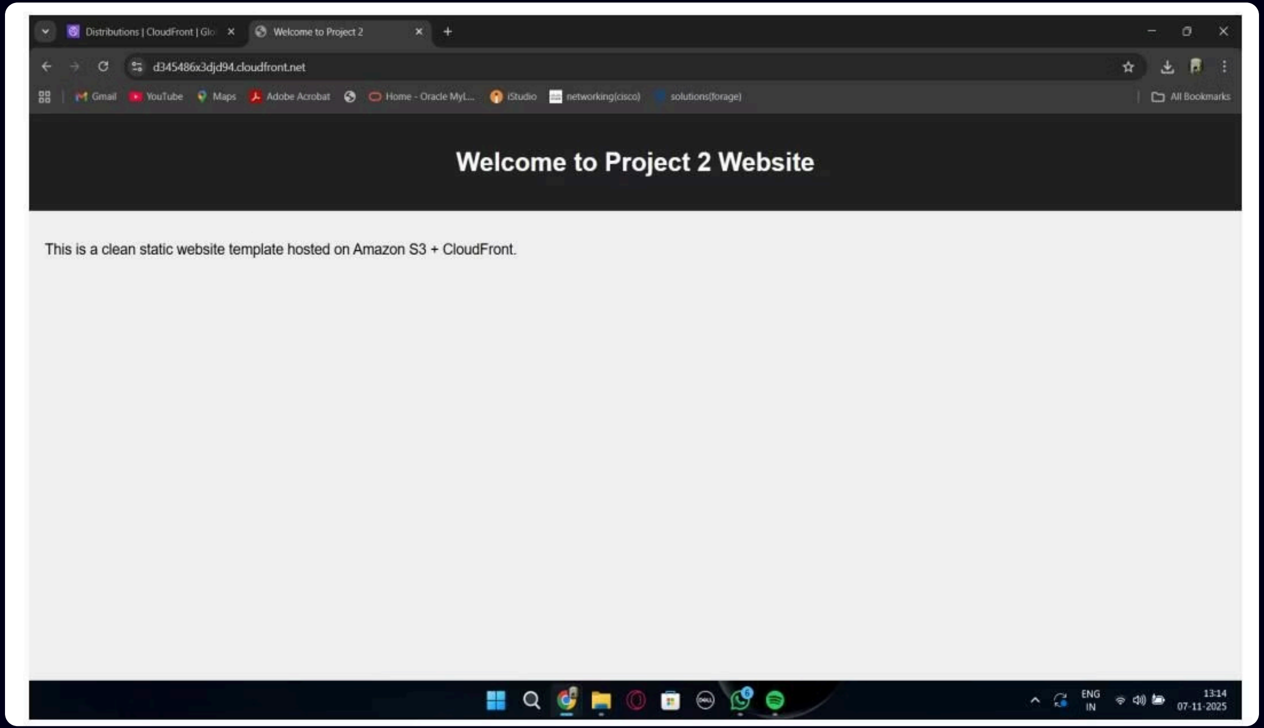


Distribution details with domain endpoint

Successful Deployment

Live Website

The website was successfully deployed and is now globally accessible through the CloudFront HTTPS URL.



Final website loaded via CloudFront URL with HTTPS

Conclusion

Serverless Excellence

Eliminates EC2 and compute costs — scalable, near-zero operational overhead

Global Performance

CloudFront accelerates delivery worldwide, reducing latency for all users

Industry Standard

Ideal for portfolios, landing pages, docs, and production-grade static apps

Why this architecture is recommended

- Low-cost hosting with no servers to manage
- Automatic scaling for traffic spikes and global access
- Reliable delivery with HTTPS, CDN caching, and strong availability

Production-ready architecture with HTTPS, global CDN caching, and near-zero cost under AWS Free Tier.

Key takeaways

This approach is simple, cost-effective, and built to scale. It removes infrastructure overhead while delivering fast, secure, and dependable access for real-world production use.

Best for teams that want a modern deployment path with minimal maintenance, strong performance, and predictable savings.